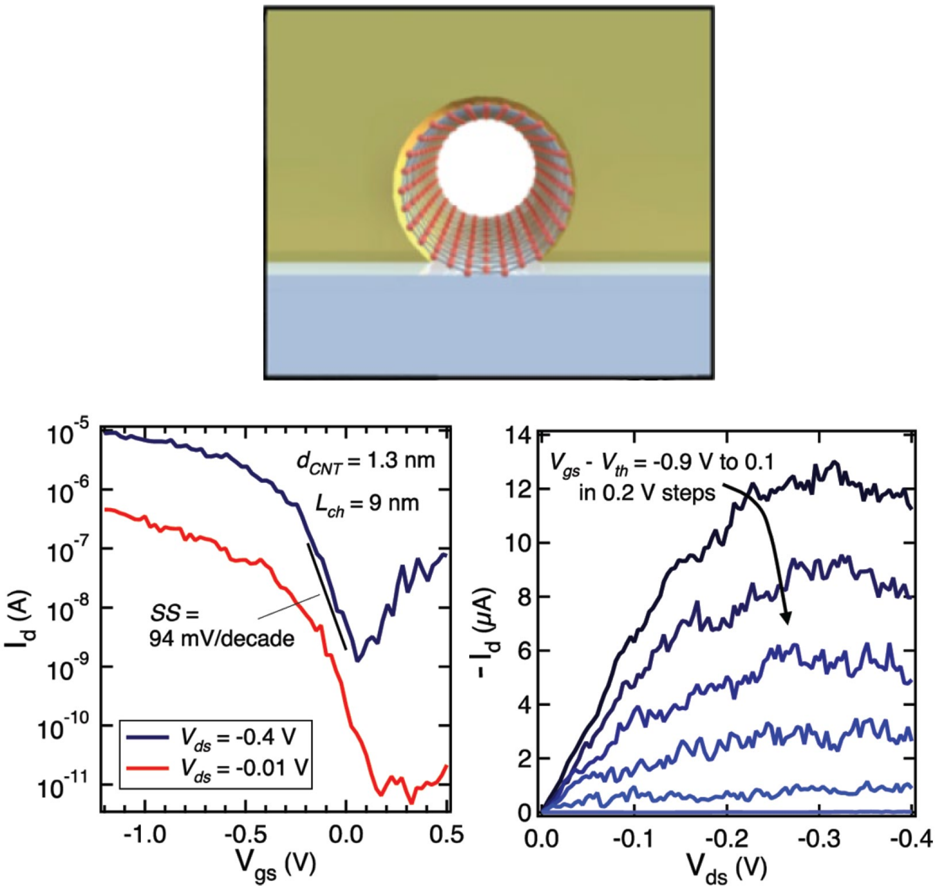

Figure 1.7: Subthreshold (off-state) and output (on-state) characteristics for a 9-nm-long channel CNTFET. Despite the low effective carrier mass there is no loss of gate control over the transport carriers. Adapted from [56]

and maximize the contact length. In the specific case of Pd and Ti contacts, the weaker nanotube-metal coupling in the case of Pd is a good explanation for the superiority of Pd-based nanotube contacts [79].

By using the transfer length model, the contact length was estimated to be inversely dependent on the contact resistance between the nanotube and the metallic pads [75, 80]. These experimental observations were initially explained by assuming dipole moments built up at the CNT–metal interface [81]. The presence of dipoles might be caused by interface states at the junctions formed by charges which provide a potential that decays within a few nanometers from the junction. A high density of dipole charge translates to a high density of interface states, which is expected to pin the Fermi level. Since in a planar contact, the dipole in a metal-CNT contact is localised in all directions, the overall effect of these interface states is to shift the bands within a few nanometers of the junction, generating a narrow barrier, whilst the energy of the bands deeper within the CNT remain the same as for a contact without interface states [82]. One conclusion one may draw from this is that by having a electrical contact which has a work function closer to the the nanotube might protect the metal–nanotube junction to undergo to Fermi level pinning effects caused by surface states.

In addition to considering different contact geometries, more advanced methodologies were applied to clarify the role of the chemical bonds produced between the metal and carbon structures, or some combination of metal-/carbon-dependent interfacial charge characteristics. Cuniberti and coworkers found that the scaling of the contact resistance is mostly driven by diagonal elements of the self-energy matrix through a comprehensive non-equilibrium Green function (NEGF) methodology [78]. The significant fluctuations of these elements for weakly interacting contact metals were not significant for contacts of infinite length, but their work demonstrates they have an important role when the contact length is finite. To quantitatively characterise the contact length scaling, they introduced the concept of an effective contact length, which is defined as a contact length for which the contact resistance is %5 greater than the infinite contact case and depends solely on the local metal-nanotube coupling rather than the tube diameter [76].

The researchers hypothesised that the most unfavourable length scaling is observed for weakly-interacting metals such as Cr, Pd, and Pt, for which the effective contact length is greater than 100 nm; on the other hand, metals such as Au, Al, and Sc were nearly insensitive to the contact length when this was decreased to 15 to 22 nm. Furthermore, the authors proposed that if the infinite length contact resistance is low, the element of the self-energy matrix must be greater than 30 meV in order to maintain a low resistance of less than 10 nm. This criterion appeared to be met by an unexpected contact material proposed by the authors, Rh, for which they projected a better scaling of the contact resistance in comparison to Au, Pt, and Pd. For long Rh-CNT connections, the scientists examined the contact resistance from Pd and Rh and found little to no difference, indicating that the scaling benefit gained by using Rh may not be worth the expenses if precise control over the nanotube anchoring location is limited.

1.4 Graphene Nanoribbons

Among the all-carbon alternatives to silicon, graphene nanoribbons have garnered considerable attention for logic applications. These require two well-defined voltage states and the ability to rapidly flip between the two states while maintaining a low leakage current in the off state. As I have already mentioned at the beginning of the chapter, one of the primary objections of graphene's use in digital electronics is the lack of an energy gap, which would allow the device to remain in an off state without any leakage current when the applied potential is low or zero [16, 23].

The confinement of large-area graphene carriers within a 1D channel creates an electronic bandgap dependent on the channel width and length [83]. Early theoretical calculations have in fact shown that when the ribbon width approached the widhts around 2-3 nm, the resulting bandgap becomes closer to those of Ge or InN. By further reducing down the width (1-2), this should technically approach bangaps closer to those of Si or GaAs [84]. In fact, early theoretical studies have demonstrated that when the ribbon width approaches widths of (3–4) nm, the resulting bandgap approaches those of Ge and InN. By further lowering the width to (3–4) nm range, this should theoretically approach the bandgaps of Si or GaAs [84].

Fig. 1.8 shows various ribbon widths and their corresponding bandgaps, as reported by theoretical and experimental studies. Moreover, this graph illustrates the paucity of ribbons with suitable widths when optimal bandgap range for CMOS applications is considered to be in the (0.1–1) eV range [16, 85]. One of the key reasons for the scarcity of appropriate GNRs is that the bottom-up process for the synthesis of graphene ribbons was still in its infancy at the time this graph was created.

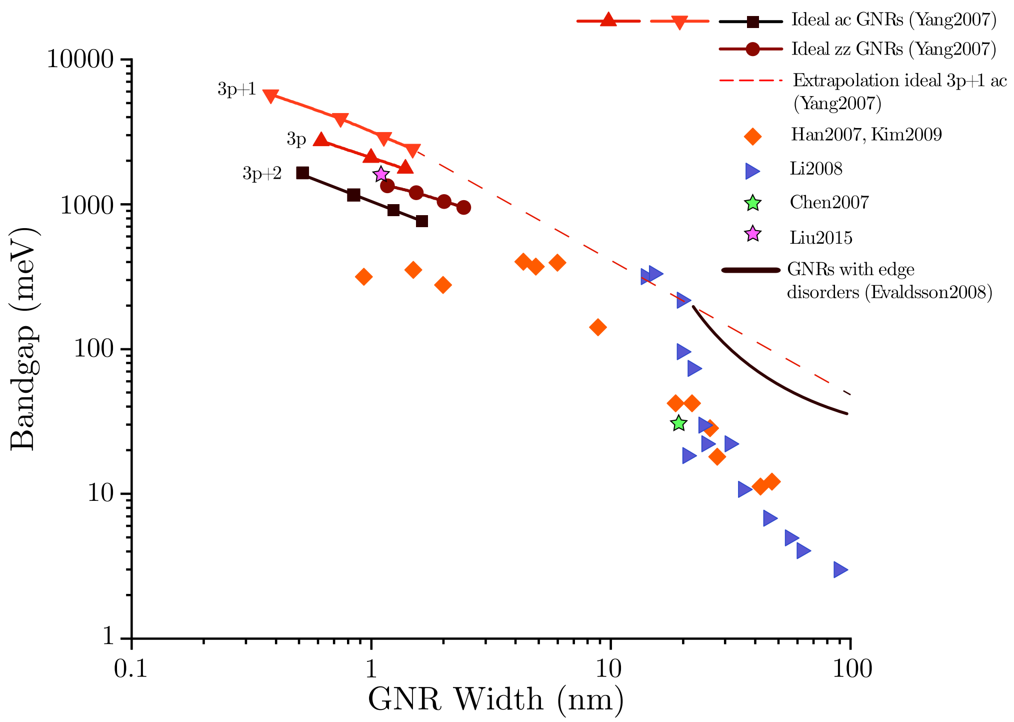

Figure 1.8: Bandgap versus nanoribbon width as reported both in experimental [**Li2008**, 83, 86, 87] and theoretical works [88, 89]. ac stands for *armchair*, while zz for *zigzag*. See main text. Adapted from [16].

The *bottom-up* methodology is a technique for synthesising graphene ribbons in which the ribbon structure is created from molecular precursors reacting using reaction patterns commonly inherited from polymer chemistry and pericyclic reactions. Pioneered by two seminal papers by Hongjie Dai [**Li2008**] and Klaus Müllen [90], it is now deemed the ideal technique to produce high-quality graphene nanoribbons.

With these class of techniques, hundred nanometer-long ribbons with widths less than 5 nm have been synthesised [91], which may be appropriate for interconnection applications requiring high current density to be maintained without causing voltage drop. Secondly, important nano-and- optoelectronics properties such as electronic bandgap, charge-carrier mobility, and optical absorption have become tunable by either varying GNRs chemical structures or by introducing opportune decorations along the ribbon edges [92, 93]. Fig.1.9 gives a glimpse on the vast variety of edge structures generated by bottom-up techniques, suggesting the effectiveness and versatility of this approach over early *top-down* approaches. These were typically accomplished by either patterning ribbons through lithography with electron-beam litography (EBL) [83, 94] or by longitudinal unzipping of single-walled nanotubes [95], or by cutting graphene sheets with metallic nanoparticles [96]. These methods, however, frequently produced wide widths and uncontrollable structural defects [89, 97]. Finally, bottom-up technology has made it possible to synthetically control the kind of defects injected into the carbon lattice, and this technological superiority is considered by many to be a substantial advantage compared to graphene and carbon nanotubes [98, 99].

1.4.1 Crystal and Band Structure

Armchair and zigzag ribbons were historically the first two edge structures studied. The variation in the edge connectivity has a significant impact on the respective band structure. In general, the energy spectrum of a GNR with length L and width w_{GNR} may be written as:

$$E_{n,m} = \pm\hbar v_F^* \sqrt{\left(\frac{n\pi}{L}\right)^2 + \left(\frac{m\pi}{w_{\text{gnr}}}\right)^2} \tag{1.22}$$

where v_F^* is the group velocity for an energy band indexed by the pair of quantum numbers n, m [100].

Steven Louie's group was one of the first to characterise the electrical properties of armchair ribbon based on the number of carbon dimer lines N across the ribbon width using tight-binding (TB) calculations [101]. The authors of this seminal study discovered that when N is a multiple of three of an integer p, the resulting ribbons feature large bandgaps that scale inversely with the width of the ribbon[†] (Tab. 1.4), even in the case the criterion $N = 3p + 1$ is met, this trend persists qualitatively. In contrast, $N = 3p + 2$ indicated that the ribbon family would be metallic. [103].

When the width of the ribbon falls below 10 nm, however, substantial electron-electron interactions induced by the system's one-dimensionality lead to significant, non-negligible self-energy correction terms that have been satisfactorily captured by

[†]The $E_g \propto 1/w_{\text{gnr}}$ relation holds when 15 nm $\leq w_{\text{gnr}} \leq$ 90 nm [102]

Table 1.4: Prediction of AGNRs electronic properties based on TB.

N	Type	E_g (eV)
$3p$	Semiconductors	$0.1 - 2.5$
$3p+1$	Semiconductors	$0.1 - 2.5$
$3p+2$	Metallic	0

Table 1.5: Prediction of AGNRs electronic properties based on the Green's function formalism within the GW approximation.

N	Type	E_g (eV)
$3p$	Semiconductors	$2.5 - 5.5$
$3p+1$	Semiconductors	$0.1 - 2.5$
$3p+2$	Semiconductors	$1 - 2$

many-body theories [102] - Tab.1.5 summarises the estimates for these corrective terms. A few years later, STM tests on surface-synthesised 5-atom wide AGNRs supported these predictions, reporting bangaps of $\simeq 100\,\mathrm{meV}$ [104]. Their findings also shed some light on the implication that finite-size effects must be considered when determining the scaling behaviour of the energy gap with ribbon length. The authors determined the orbital energies with respect to the Fermi level by measuring the differential conductance for ribbons of various lengths. As anticipated, the gap between the Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO) rapidly reduces as ribbon length increases. What they also report in their work is that while the energy gap scales quadratically with ribbon length $E_g \propto 1/L^2$, the energy spacing between electronic states that belong to the same band scales inversely with ribbon length $E_g \propto 1/L$, as predicted by modelling the behaviour of these states using energy states of graphene with linear dispersion. In addition, between 4- and 5-monomer long ribbons, the HOMO level passes the Fermi level, causing the ribbon to become positively charged. There is no rapid change in the orbital energies due to the ribbons' charge, indicating that the charging energy is negligible and may be disregarded [105].

ZGNRs's electronic properties are qualitatively and quantitatively distinct from those of AGNR. In order to define the unit cell of a ZGNR, just two atoms per unit

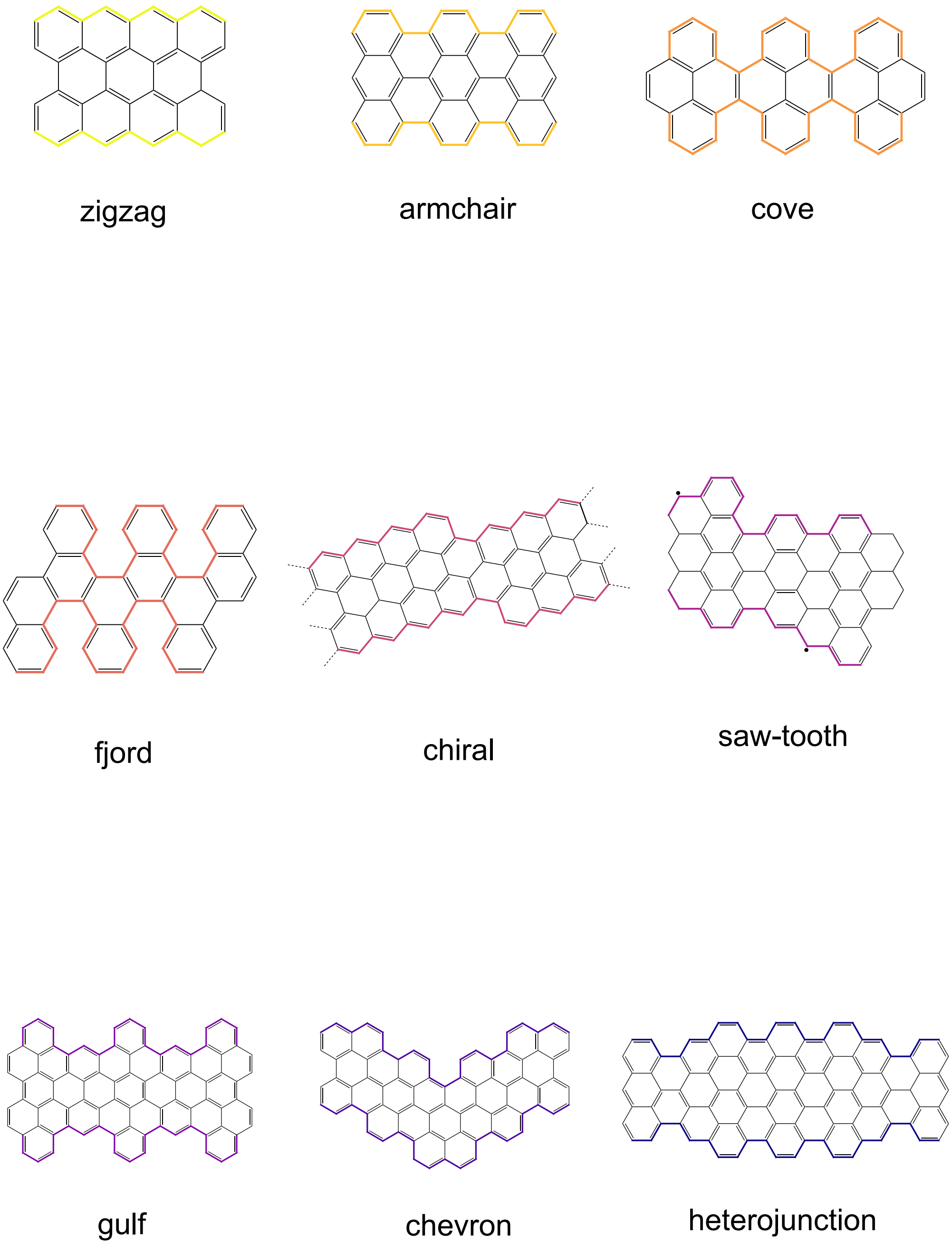

Figure 1.9: Different geometries and edge structures of graphene nanoribbons.

cell are required, as opposed to four for an AGNR. TThe locations of these two atoms within the unit cell and the translation symmetry of the lattice give rise to a symmetry element–a 2_1 screw axis–present along the direction of width extension. Most notably, the presence of glide planes and screw axes in nonsymmorphic space groups‡ (as for the wave vector group) causes energy bands to converge along certain high symmetry points and axes in the Brillouin zone [106]. This *flat band* region is positioned at wave vectors $2\pi/3 \leq k \leq \pi$, and the electronic states in this region have been interpreted as localised states along the zigzag edge by looking at charge distribution maps [107]. In one-dimensional systems, these electronic states are known as *edge states*, and their contribution to the entire wavefunction becomes minimal when their widths approach the micrometre range, other than exhibiting an exponential decay of their DOS with distance from the ribbon's centre [101, 108].

When the wave vector matches the ratio between π and the unit cell distance d_z, i.e. $k = \pi/d_z$ (Fig. 1.10**c**), it is possible to induce an energy gap between energy valleys. It has been proved by spin-polarized ab-initio calculations that two distinct magnetic orderings are distinguishable in the ground state.: a ferromagnetic (↑↑, FM) configuration for spins aligned along a single edge and an antiferromagnetic (↓↓, AFM) condition for spins belonging to two distinct edges [109]. *Half metallicity* is the property of experiencing the phase transition from an AFM configuration (semiconducting) to an FM state (metallic), and it has been established that this property should be theoretically accessible if a magnetic field is induced parallel to the ribbon extension plane [110].

1.4.2 Edge States

A few years later, the chiral DOS for the electronic states at the ribbon edges had to be incorporated into the interpretation of the previous mentioned localised states and the exotic features that followed these [111]. Chirality is a remarkable phenomenon that arises when the charge carriers' time-reversal symmetry is broken,

‡Nonsymmorphic groups are space groups that are not uniquely defined by the lattice translations' repeating of a point-group motif. Te and α-quartz are examples of nonsymmorphic groups.

and it is possible to discriminate between carrier wavefunctions evolving in time with a clockwise phase and those evolving with a counterclockwise phase. In the molecular domain, geometrical chirality refers to any molecule belonging to a symmetry group without improper rotations, i.e. the spatial symmetry is such that the rotamer cannot be superimposed on its mirror image [112].

Similary to carbon nanotubes, graphene nanoribbons can have chiral edges. Typically, chirality may be described by a chiral angle θ which may be defined as:

$$\theta = \arcsin \sqrt{\frac{3}{4}\left(\frac{m^2}{n^2 + nm + m^2}\right)} \tag{1.23}$$

where the nanoribbon's edge orientation is determined by translation vector (n, m) of the graphene lattice. Zigzag and armchair are characterised by $\theta = 0°$ and $\theta = 30°$, respectively, while chiral edges have θ in the range between. Steven Louie and co workers employed a mean-field Hubbard Hamiltonian for modelling the GNR dispersion relation, assuming the on-site Coulomb repulsion U comparable to the hopping integral t between two adjacent sites ($U/t \sim 1$):

$$\begin{aligned}\mathcal{H}_{\mathrm{MF}} = &-t \sum_{\langle ij\rangle,\ \sigma} \left(a^{\dagger}_{\sigma,i} a_{\sigma,j} + \mathrm{H.c.}\right) \\ &+ U \sum_{i} \left(n_{i\uparrow}\langle n_{i\downarrow}\rangle + \langle n_{i\uparrow}\rangle n_{i\downarrow} - \langle n_{i\uparrow}\rangle\langle n_{i\downarrow}\rangle\right),\end{aligned} \tag{1.24}$$

whereby the second summation is performed on spin-polarized densities $n_{i\sigma} = a^{\dagger}_{i\sigma} a_{i\sigma}$. This Hamiltonian preserves electron-hole symmetry, which means that the density of states is symmetric with respect to the Fermi level.

The experimental evidence was accurately characterised by the wavefunction solutions of the associated eigenvalue problem, which predicted two pairs of peaks divided in the DOS and linked to two distinct magnetic orderings. One peak split was caused by the AFM correlation between opposite edges, whereas the second, greater peak split was caused by FM ordering along a single edge (Fig. 1.10). The energy difference between the two magnetic orderings was estimated to be in the order of tens of meV per edge atom and to decrease as the ribbon width increases, eventually becoming negligible when the width is significantly larger than the

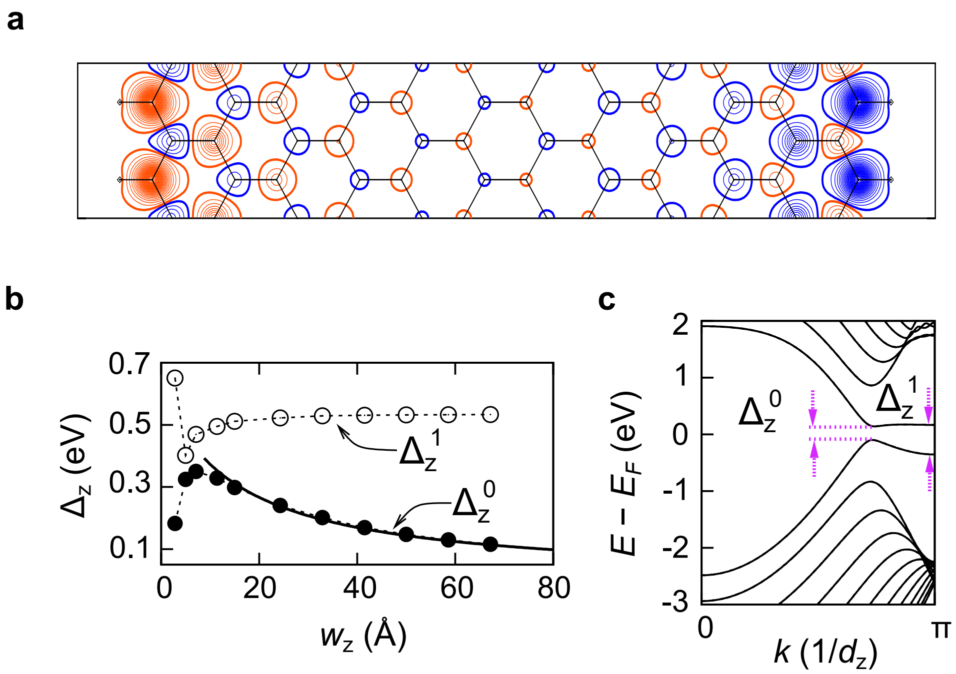

Figure 1.10: (a) Contour plot representing the difference between spin-polarised ab-initio DOS of a 12-ZGNR. The high density of $\rho_\alpha(r)$ red lines ($\rho_\beta(r)$, blue line) to the left backbone edge demonstrates the delocalisation of edge states. Blue (red) line represent (b) Direct bandgap Δ_z^0 and the energy splitting Δ_z^1 as a function of N_z-ZGNR width w_z. For $N_z \geq 8$, the fit is shown as a solid line, whereas dotted lines are guides. (c) 12-ZGNR bandstructure, plotted as a function of the normalised wavevector. Adapted from [101].

exponential decay length of spin-polarized states [113]. In theory, it should then be possible to probe this energy difference by either study the tunnelling density of states or by using high-frequency resonante techniques.

These theoretical predictions have been supported by experimental data after many years of effort. Using electron-spin resonance techniques at a frequency of 94 GHz, Slota and colleagues elucidated magnetic orderings that were localised along the edges [114]. Their ribbon structure was distinguished by the presence of stable nitroxyl radicals as spin injectors along the ribbon's edges, permitting the separation of the magnetic edge fingerprint from the signal of the radical. In addition, their study shown that the edge state has a longer coherence time than the majority of π-extended organic systems, even at room temperature and in powder, which makes graphene nanoribbons particularly interesting for spintronics applications. Therefore, it would be desirable to investigate the tunability of magnetic characteristics by electrical control in single zigzag graphene nanoribbons, leveraging experimental techniques borrowed from molecular electronics, such as tunnel break junctions, for instance. In fact, this was one of the present thesis's primary objectives.

1.4.3 Graphene Nanoribbons Field-Effect Transistors

One of the first graphene nanoribbon field-effect transistors (GNRFETs) prototype was realised by the Dai's group by synthesising through sonochemistry. Source and drain were fabricated using Pd, which is a metal with a high work function and thus minimizes the Schottky barrier height for thr holes when a *p*-type transistor

The Dai group developed one of the first bottom-up GNR employing sonochemistry and Pd for ribbon electrical contact. As discussed previously, the large work function of Pd should reduce the Schottky barrier height for minority carrier injection in doped substrates [115]. The authors examined the impact of ribbon widths as small as $w_{\mathrm{GNR}} \simeq 2.0(5)\,\mathrm{nm}$ on the electrical performance of the material while keeping ribbon lengths roughly constant $l_{\mathrm{GNR}} \simeq 236$. Although a few impressive transistor key figures, were reported, degraded carrier mobility within the $\simeq$ 50–200 $\mathrm{cm^2V^{-1}s^{-1}}$ range were also seen for narrower GNRs.

The wider bandgaps, the superior quality of edge structures, a thin gate oxide, and a short GNR channel were among the characteristics cited by the author as essential for achieving increases of approximately 10^5 for the $I_{\mathrm{on}}/I_{\mathrm{off}}$ at room temperature, as well as high current density and transconductance. The team also compared the performance of their device to that of Pd-contacted carbon-nanotube field-effect transistor (CNFET) on 10 nm SiO_2 with comparable channel lengths and variable tube diameters.

They discovered that the tube diameter was crucial in determining the current density delivered in the ON state and the leakage current in the OFF state, arguing that the negatively correlated role of a large positive SB and a short mean free path of the carriers limited by acoustic phonons and defects seemed to be the main responsible factors [116, 117].

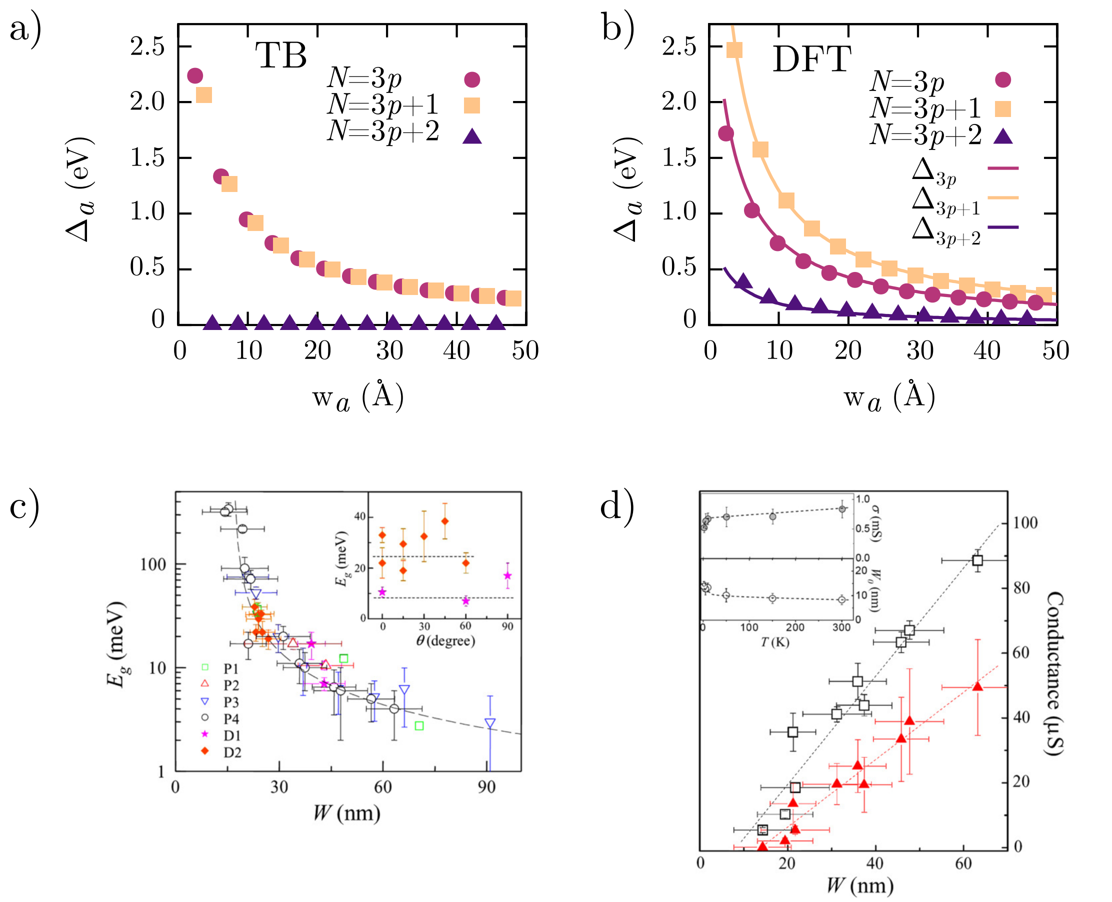

Figure 1.11: **a,** Bandgaps Δ_a calculated at TB level for three families of AGNRs as a function of the GNR width w. **b,** Same as in **a,** but with DFT-level accuracy. **c,** Experimental trend for the bandgaps as a function of ribbon's width. The authors tried different Crystallographic orientation for the ribbon founding no dependence of the direction, as contrary expected from early predictions. **d,** Conductance vs width of parallel GNRs measured at a gate voltage $V - V_{\mathrm{G}} = 50\,\mathrm{V}$. Squares are values measure at 300 K whereas triangle corresponds to 1.6 K. The insets show the conductivity (upper) and the inactive GNR width (lower) obtained from the slope and x intercept of the linear fit at varying temperatures. Adapted from [83].

Despite a few remarkable transistor key numbers such as $I_{\mathrm{on}} \simeq 2000\ \mu\mathrm{A}\,\mu\mathrm{m}^{-1}$, $I_{\mathrm{on}}/I_{\mathrm{off}} \simeq 10^6$, $SS \simeq 210\,\mathrm{mV\ dec}^{-1}$, and $g_{\mathrm{m}} = 900\ \mu\mathrm{S}\,\mu\mathrm{m}^{-1}$, reduced carrier mobility within the $\mu \simeq 50$–$200\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ range was also observed for narrower ribbon. The rationale for this unexpected outcome is twofold. To begin, it is well documented in traditional solid-state semiconductors that diminishing mobilities lead to increased bandgaps [118]. Second, the authors suggested that edge effects dominate carrier transport in narrow ribbons. Wang and co workers proposed a phenomenological model in which edges, defects, and phonons were regarded as independent factors in order to quantify the contribution of edges to the overall carrier mean free path ℓ [115]:

$$\frac{1}{\ell} = \frac{1}{\ell_{\text{edge}}} + \frac{1}{\ell_{\text{phonon}}} + \frac{1}{\ell_{\text{defect}}} \tag{1.25}$$

The likelihood of a carrier scattering event occuring is then estimated to be inversely proportional to the free mean path:

$$\ell_{\text{edge}} = \frac{k_{\parallel} w_{\text{gnr}}}{P k_{\perp}} \tag{1.26}$$

As we saw in the previous section, for ribbon widths less than 20 nanometers, it is possible to approximate the ribbon's energy gap as inversely proportional to the ribbon's width, provided electron-electron correlations are neglected [119]:

$$E_g \simeq \frac{\hbar v_F}{w_{\text{gnr}}} \tag{1.27}$$

By inserting Eq. 1.27 and the dispersion relation E_k one into Eq. 1.26, an expression for the dependence on the width of the mean free path may be written as:

$$\ell_{\text{edge}} = \frac{w}{P}\sqrt{\left(1 + \frac{E_k}{E_g}\right)^2 - 1} \tag{1.28}$$

According to Eq. 1.28, an increase in width and/or an improvement in edge quality should lead to a rise in the mean free path. These hypotheses were proven experimentally, as wider sub-10 nm ribbons with greater mobilities were measured [85, 115].

Table 1.6: Comparison amongst different FET realised with graphene nanoribbon as conducting channel. Top and bottom.

	Structural Parameter			Device Parameter	
Fabrication Approach	Edge Structure	Width (nm)	E_g (meV)	$I_{\mathrm{on}}/I_{\mathrm{off}}$	Mobility ($\mathrm{cm^2V^{-1}s^{-1}}$)
Top-Down	—	5	145	47; 10^5 (5.4 K)[120]	—
	—	5	489.4 ± 19.0	$> 10^4$ (290 K)[121]	—
	—	15	120 ± 23	—[121]	—
	Mixed	10	140	10^5 (4 K)[122]	—
	—	12	100	10; 10^6 (4 K)[123]	—
	—	sub-10	200	—[55‡]	—
	—	10 – 20	10 – 15	4.5; 10 (20 K)[124]	—
	Closed	2.8	494	10^5	2443_{μ_h}
	Closed[125]	9.9	$891/[\mathrm{w_{gnr}} - 1.0]$	$< 10^2$	3776_{μ_h}
	—	100 – 400	18 – 1	20 – 3.5[126]	—
Bottom-up	Cove	0.69 – 1.13	1,880	$>^{\dagger} 10^{10}$	
	AGNR	0.95	2,100	$^{\dagger}10^5$	
	Chevron	1.73	$3,100 \pm 400$	—	
	—	2	—	$> 10^6$	
	AGNR	0.63	$1,690 \pm 100$	$>^{\dagger} 10^{1}0$	
	AGNR	0.95	$1,690 \pm 70$	—	
	AGNR	1.27	$1,130 \pm 50$	$^{\dagger}10^9$	
	AGNR	1.37	$1,600 \pm 400$	—	
	AGNR	1.58	$1,030 \pm 40$	$^{\dagger}10^8$	
	AGNR	5 – 9	≤ 400	$10^5 - 10^6$	
	ZGNR	6	1,400	—	
Epitaxial	—	900	—	—	
	AGNR	2 – 40	> 500	—	
	—	40	—	10; > 25 (4 K)	

The experimental validation of the theoretically predicted relationship between ribbon width and energy gap occurred years. In 2007, Philip Kim's team demonstrated that the theoretical description lacked a few characteristics that were apparent in experiments by using micrometer-long, sub-100-nanometer-wide graphene nanoribbons produced by lithograpic means [83].

The experiments they perfomed suggested that even if the transport could–in principle–be ballistic disorder effects may produce charge localisations, and this effect becomes greater as the ribbon's length increases [127–129]. This hypothesis was later confirmed by a detailed analysis conducted by the Manchester group, which demonstrated that the electron–electron interactions between electrons within the dots are not negligible when the ribbon's width is reduced and that the conductance was suppressed due to Coulomb blockade, which also explained the deviation of larger energy gaps for small ribbons compared to theoretical studies [128, 130].

In an attempt to empirically resolve this conundrum, Kim's group suggests that the explanation may be found in a combination between those works. Hopping charge transfer mechanism through localised states of comparable size to the ribbon's width would result in a transport gap as opposed to a true energy gap [97]. The empirically measured bandgap, i.e. the gate voltage region ΔV_{G} where conductance is suppressed, may then be related to the single-particle energy gap via [6]:

$$E_g = \hbar v_F \sqrt{2\pi C_{\mathrm{G}} \Delta V_{\mathrm{G}}/|e|} \tag{1.29}$$

where C_{G} is the capacitive coupling between the ribbon and the gate.

In addition, the authors measured the temperature dependence of the conductance and reported a strong suppression of the conductance minimum G_{min}, which suggests a thermally activated process is involved in the formation of the energy barrier between the valence and conduction bands, contrary to what would be predicted according to the Coulomb blockade model [131]. Even with different ribbon widths, the scientists discovered a linear scaling of the conductance $G = \sigma(w_{\mathrm{GNR}} - w_0)/L$ when this was normalised across an active transport region $(w_{\mathrm{GNR}} - w_0)$, being $\sigma \simeq 1.7\,\mathrm{mS}$ the sheet conductivity. Comparing similar trends reported for SB-restricted junctions led to the conclusion that the graphene nanoribbons-metal connection may have been limited by the formation of a Schottky barrier that dominates the charge injection [132].

Utilizing pure carbon nanotubes and compressing them under high pressure to make graphene nanoribbons has recently yielded excellent device performance [125]. The nanotube samples were loaded and sealed in a sample chamber positioned in the centre of a pre-indented tungsten gasket, which was then compressed between two diamond anvils from ambient pressure to 22.8 GPa while the structural evolution was monitored by in-situ Raman spectroscopy through the diamond window. The authors observed a resonant radial breathing mode peak at 116 cm^{-1}, corresponding to a nanotube with a diameter $\simeq$ 2.1 nm, dropping in peak intensity as the pressure climbed, and finally becoming practically invisible when the pressure surpassed 5.5 GPa. The peak position shift and absence of the radial breathing mode was attributed to an irreversible structural transition in nanotubes that occurs at high pressure and affects their sub-band energy gaps.

The authors found a remarkable on/off ratio of approximately four orders of magnitude at extremely low bias. This was criticised by the scientific community, as ratios in this range for such modest biases have only previously been reported for carbon nanotube transistors and never for graphene nanoribbon devices (Tab. 1.6). In addition to this criticism, the authors said that they did not observe any radial-breathing-like mode. Previous research on seven- and nine-atom-wide armchair GNRs generated by on-surface synthesis indeed revealed a low-frequency radial-breathing-like mode in the Raman spectra using 532 and 785 nm excitation lasers, respectively. [133].

The device they fabricated was a standard back-gated silicon dioxide type of the device which showed p-type on state conductivity of about:

$$\sigma = \frac{G_{\mathrm{sd}} L_{\mathrm{ch}}}{w_{\mathrm{gnr}}} = 7.42 \text{ mS} \tag{1.30}$$

for a channel of width of ($w_{\mathrm{gnr}} \simeq 2.8$ nm). They further test the relationship between the carrier mobilities for increasingly wider ribbon's widths up to 10 nm.

The experiments reported hole mobilities† of about $\mu \simeq 2.443\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ for the 2.8 nm wide and $\simeq 3.776\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ for the 10 nm wide, which are currently among the highest values observed, despite possible uncontrolled scattering events of the carriers with the non-spherical bulb-like ends cross at the two sides.

Another issue in creating reproducible devices is transferring ribbon onto a substrate to measure its electrical characteristics. Because stamping methods that rely on mechanical contact of the ribbon growth and transfer substrate frequently result in poor-performing or irreproducible GNRFETs [90], Michael Crommie's group developed a robust transfer employing a stack of PMMA/GNR/Au/Mica [134]. Poly-methyl methacrylate (PMMA) was first spun-cast onto GNRs and baked to generate a PMMA/GNR/Au/Mica stack, which was then floated on concentrated HF (40%), causing the mica substrate to delaminate from the Au growth layer. The leftover PMMA/GNR/Au film stack was then rinsed twice in water before being put to Au etchant. Finally, the PMMA/GNR film was washed and drawn onto a silicon substrate before being baked to eliminate any remaining water. Finally, acetone was used to remove PMMA, leaving GNRs on an insulating surface.

At $V_{\mathrm{SD}} = 1\,\mathrm{V}$, the resultant transistor displayed gate-modulated conductance with on-currents ranging from tens of pA to a few nA. The authors reported that the electrical transport was mostly dominated by a significant Schottky barrier $\phi_{\mathrm{B}} \simeq 1.25\,\mathrm{eV}$, which was caused by adopting a much narrower ribbon with a wider bandgap. They also reported that, despite the fact that the contact between the ribbons and Pd must have been only a few to perhaps ten nanometers long, they were able to measure $I_{\mathrm{on}}/I_{\mathrm{off}} \simeq 10^3$ at $V_{\mathrm{SD}} = 1\,\mathrm{V}$, even if experiment on nanotube transistors have demonstrated resistance increases as contact length is reduced past the electron mean free path†. The off-state leakage was attributed to the holes tunneling through the drain barrier, and therefore being largely temperature independent but strongly sensitive on the bias, as larger ones will significantly shrink the width of the Schottky barrier.

†Calculated according to standard FET theory $\mu = \frac{g_{\mathrm{m}} L_{\mathrm{ch}}}{C_{\mathrm{G}} V_{\mathrm{SD}}}$ [33]

†In graphene and CNTs this value is often referenced at $\ell \simeq 200\,\mathrm{m}$ [75, 135]. See also Sec. 1.3

The authors also suggested several points to ameliorate their device's criticalities. Lowering the contact work function would reduce the source conduction band barrier height and correspondingly increase the drain valence band barrier, resulting in both improved on- and off-state performance. Further improvement in band alignment should also arise through the use of wider GNRs with bangaps around $\simeq 1.4\,\mathrm{eV}$, as lower effective mass and Schottky barriers are expected for smaller bandgaps. Lastly, the researchers envisioned that longer GNRs [136] might also reduce the contact resistance by increasing L_c.

The authors also proposed a certain number of measures to mitigate the device's shortcomings. Reducing the contact work function ($\phi^{\mathrm{Pd}} \simeq 5.2$–$5.6\,\mathrm{eV}$) would decrease the barrier height of the source conduction band and increase the barrier height of the drain valence band, resulting in enhanced on- and off-state performance. As effective mass and Schottky barriers are projected to be lower for smaller bandgaps, the usage of wider GNRs with bandgaps $E_g \simeq 1.4\,\mathrm{eV}$ could also result in a further enhancement of band alignment. In conclusion, the researchers hypothesised that longer GNRs could also lower contact resistance by increasing L_c.

Using high-k dielectrics is another viable method for enhancing the properties of transistors. The usage of such dielectrics is necessitated by the fact that, when scaling down the geometrical dimensions of transistors, the oxide thickness cannot be scaled in the same manner as the other geometrical parameters, as this would result in leakage currents through the oxide. This phenomenon could also result in a field that exceeds the breakdown field of silicon dioxide. By raising the permittivity of the dielectric, one can afford a sicthicker dielectric while still achieving the same capacitance as with a thinner silicon dioxide dielectric.

Llinas and colleagues investigated the output characteristics of GNRFETs with widths of nine and thirteen atoms that were fabricated using two distinct gate dielectrics [137]. They compared the output response of a typical 50 nm-thick back-gated silicon dioxide substrate to that of a high-κ dielectric local gate (HfO_2, EOT$\simeq$1.5 nm). For the latter device at room temperature, current densities

$I_{\mathrm{on}} > 1\,\mu\mathrm{A}$ measured at $V_{\mathrm{SD}} = -1\,\mathrm{V}$ with the narrower ribbon were amongst the highest ever reported, combined with an excellent $I_{\mathrm{on}}/I_{\mathrm{off}} \simeq 10^5$.

Through a temperature-dependent experiment, however, the authors noted that Schottky barriers generated at the contacts again dominated transport. While the high-κ dielectric lowers the carrier tunneling, the Au/Pd–GNR junction leads to the formation of Schottky junctions partly due to the mismatch between the work functions of the metal and the ribbon. In order to lessen the mismatch and prevent the Schottky-dominated transport regime, adopting graphene as a contacting material could be one feasible approach [138, 139].

1.5 Concluding Remarks

Graphene nanoribbons exhibit intriguing physical and chemical properties not only for technological advancement in areas where graphene and carbon nanotubes lack performance, but also for the physics they can reveal. At this time, bottom-up graphene nanoribbons have demonstrated remarkable transport and exotic properties, but these have been studied solely on the synthesis substrate. Attempting to investigate these properties at the device level will be one of the next major obstacles in the nanoribbon field. At this time, these evidences are largely unavailable and unclear.

There are three roles to which graphene nanoribbons may technologically contribute. The first application is that of sensors. Given the ability to functionalize the edges with a variety of chemical substituents, a large number of studies in various gas atmospheres have demonstrated excellent selectivities and good flexibility. A second papable field could be the integration in low-power devices, where by reducing required supply voltages, ribbons with lower bandgaps and consequently higher mobilities could be produced. A third and more enthusiastic application would be the development of dechorence-tolerant quantum hardware utilising their topological properties.